FROM S_2 TO S_4: NEHARI'S THEOREM ON THE PALEY–WIENER SPACE OF THE DISC

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Abstract. For Hankel operators on the Paley–Wiener space of the disc, Nehari's theorem holds in the Schatten class S_2 and fails in S_p for $p > 4$. We discuss the endpoint S_4. We recall the proof in S_2, which rests on a single bounded function, $e^{4\pi i|x|}$, and read it in terms of wave packets: it says that the complete bush of packets through a point, over all depths, is bounded. We then write the S_4 problem in the same language, explain why 4 is the natural threshold, prove the geometric ingredients that are available, and isolate the estimate that remains. The S_2 argument is quadratic and translation invariant; the S_4 problem asks for a bounded bush adapted to the function being tested.

1. Introduction

Let D be the unit disc in $\mathbb{R}^2$ and let $\mathrm{PW}^p(\Omega)$ denote the functions in $L^p(\mathbb{R}^2)$ whose Fourier transform, $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x\cdot\xi}\,\mathrm{d}x$, is supported in $\overline{\Omega}$. A Hankel operator on $\mathrm{PW}^2(D)$ is the bilinear form $H_\varphi(f,g) = \langle fg, \varphi\rangle$; on the Fourier side it is the operator on $L^2(D)$ with kernel $\widehat{\varphi}(\xi+\eta)$. Only the restriction of $\widehat{\varphi}$ to $2D$ matters, so φ is a symbol of H_φ whenever its Fourier transform agrees with the given one on $2D$; *Nehari's theorem holds in* S_p if every $H_\varphi \in S_p$ has a bounded symbol. By the Hahn–Banach theorem this is equivalent to

$$(\mathrm{N}_p) \qquad |\langle h, \varphi\rangle| \le C\|h\|_1\|H_\varphi\|_{S_p}, \qquad h \in \mathrm{PW}^1(2D),$$

for, say, Schwartz h and φ with Fourier support in $2D$. Since $S_p \subset S_q$ for $p < q$, the set of p for which Nehari's theorem holds is an interval.

History. For the Hardy space of the circle, Nehari [9] proved that every bounded Hankel operator has a bounded symbol, and Peller [11] identified the Hankel operators in S_p with Besov spaces, which lie in L^∞; so on the circle Nehari's theorem holds in every S_p. For multiplicative Hankel forms (the infinite polydisc), Ortega-Cerdà and Seip [10] showed that Nehari's theorem fails for bounded forms, Helson's inequality gives it in S_2, Brevig and Perfekt [7] showed that it fails in S_p for $p > 5.7388$, and Bampouras [2] recently lowered this to $p > 4$ and singled out the endpoint $p = 4$.

For the disc, Brevig and Perfekt [8] showed that Nehari's theorem fails for bounded Hankel operators. Bampouras and Perfekt [4] described the Hankel operators of class S_p by Besov spaces adapted to the boundary of $2D$ for $p < 4$, with the necessary direction valid for all p [4, Theorem 4.3]; this gives Nehari's theorem in S_1. In [6] they proved it in S_2 through a Paley–Wiener version of Helson's inequality, and Bampouras [3] gave a short proof of that inequality for all convex sets free of affine lines. For polygons Bampouras and Perfekt [5] proved Nehari's theorem in every S_p, $p < \infty$. On the other hand Bampouras [1] proved that it fails in S_p, $p > 4$, for every convex domain with a curved boundary piece, in particular for the disc. So for the disc the interval contains $[1, 2]$ and is contained in $[1, 4]$.

Problem 1.1. Does Nehari's theorem hold in $S_4(\mathrm{PW}^2(D))$?

A positive answer gives every $p \le 4$. We believe the answer is positive, for the reasons in Section 3. Section 2 recalls the S_2 proof and reads it in terms of wave packets. Section 3 states the S_4 problem in the same terms and explains the threshold. Section 4 gives equivalent and sufficient formulations, Section 5 the tools that use the disc, and Section 6 what a proof must

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overcome. Only results with complete proofs are stated as propositions or lemmas; heuristic statements are marked as such.

Notation. Write $\omega = \chi_D * \chi_D$, so $\omega(\xi) = |D \cap (\xi - D)|$. For dyadic depths $\delta = 4^{-k}$ put $L = \delta^{-1}$ and $W = \delta^{-1/2}$, so $W^2 = L$. For each depth fix a maximal $\sqrt{\delta}$-separated set of directions (unit vectors up to sign); an arc of length ℓ contains at most $\ell/\sqrt{\delta} + 1$ of them. A parabolic Whitney box Q of depth δ is a curved rectangle $\{\xi \in 2D : \delta \le 2 - |\xi| < 4\delta,\ |\arg \xi - \theta| < \sqrt{\delta}\}$ with θ one of these directions; it has area $V_Q \asymp \delta^{3/2}$, and $\omega \asymp V_Q$ on Q. A tube of depth δ and direction θ is a rectangle of length L along θ and width W; for each direction we fix a tiling of the plane by such tubes. Tubes of depth δ are the physical dual of the boxes of depth δ, and a wave packet ψ_T is an L^∞-normalized modulated bump adapted to T, with $\widehat{\psi_T}$ supported in a slight enlargement of the corresponding box. The packet mass of h on T is $m_T(h) = |\langle h, \psi_T \rangle|$. We write Out for the annihilator of $\mathrm{PW}^1(2D)$ in L^∞, the bounded functions with spectrum outside the open disc $2D$; by duality,

$$\sup_{\|h\|_1 \le 1} |\langle h, \Phi \rangle| = \operatorname{dist}_{L^\infty}(\Phi, \mathrm{Out}). \tag{1}$$

2. The case S_2

Since $\|H_\varphi\|_{S_2}^2 = \int_{2D} |\widehat{\varphi}|^2 \omega$, the Cauchy–Schwarz inequality shows that (N_2) is equivalent to Helson's inequality

$$\int_{2D} \frac{|\widehat{h}(\xi)|^2}{\omega(\xi)}\, \mathrm{d}\xi \le C \|h\|_1^2, \qquad h \in \mathrm{PW}^1(2D). \tag{2}$$

Bampouras [3, Theorem 2.1] proved (2) for all convex sets free of affine lines by a short argument: $1/\omega$ is dominated by the Fourier transform of a bounded function. For the disc everything is explicit.

Proposition 2.1 (Bampouras)**.** *Inequality* (2) *holds with* $C = 6\pi$.

Proof. For $\varepsilon > 0$ let $\Psi_\varepsilon(x) = -2\pi^2 e^{4\pi i|x| - 2\pi\varepsilon|x|}$. In polar coordinates, with $\int_0^\infty r e^{2\pi i r z}\, \mathrm{d}r = -(2\pi z)^{-2}$ for $\operatorname{Im} z > 0$,

$$\widehat{\Psi_\varepsilon}(\xi) = \frac{1}{2}\int_0^{2\pi} \frac{\mathrm{d}t}{(2 - \xi\cdot\theta_t + i\varepsilon)^2} \longrightarrow \frac{1}{2}\int_0^{2\pi} \frac{\mathrm{d}t}{(2 - \xi\cdot\theta_t)^2} = \frac{2\pi}{(4 - |\xi|^2)^{3/2}},$$

where $\theta_t = (\cos t, \sin t)$, uniformly on compact subsets of $2D$. Next, $\omega(\xi) \ge (4 - |\xi|^2)^{3/2}/6$: with $|\xi| = 2\cos t$, $0 \le t \le \pi/2$, we have $\omega = 2t - \sin 2t$ and $(4 - |\xi|^2)^{3/2} = 8\sin^3 t$, and $f(t) = 2t - \sin 2t - (4/3)\sin^3 t$ has $f(0) = 0$ and $f'(t) = 4\sin^2 t(1 - \cos t) \ge 0$. Let $\widehat{h} \in C_c^\infty(2D)$. Then

$$\int \frac{|\widehat{h}|^2}{\omega} \le 3\pi^{-1} \int |\widehat{h}|^2 \frac{2\pi}{(4 - |\xi|^2)^{3/2}} = 3\pi^{-1} \lim_{\varepsilon\to 0} \langle h * \Psi_\varepsilon, h\rangle \le 3\pi^{-1}\cdot 2\pi^2 \|h\|_1^2.$$

Such h are dense in $\mathrm{PW}^1(2D)$ [6], and Fatou's lemma finishes the proof. □

The function $h_0 = \check{\chi}_D^2$ has $\widehat{h}_0 = \omega$ and $\|h_0\|_1 = |D|$, so the best constant in (2) lies in $[1, 6\pi]$.

Remark 2.1 (The complete bush is bounded). The proof says that the quadratic form $\int |\widehat{h}|^2/\omega$ is dominated by $\langle h * \Psi, h\rangle$ with $|\Psi|$ constant. In terms of packets, heuristically: $1/\omega \asymp \sum_Q V_Q^{-1} m_Q$ for a smooth Whitney partition $\sum_Q m_Q = 1$, and $V_Q^{-1}\mathcal{F}^{-1} m_Q$ is a packet adapted to the tube through the origin dual to Q. So $e^{4\pi i|x|}$ is, up to constants and smooth factors, the sum of all L^∞-normalized packets through the origin, over every direction and every depth, and (2) reads

$$\sum_T m_T(h)^2 \lesssim \|h\|_1^2 \quad \Longleftarrow \quad \Big|\sum_{T \ni 0} \psi_T\Big| \lesssim 1.$$

At a single depth the bush through a point has size W at its centre (Section 5); summed over all depths with the phases of $e^{4\pi i|x|}$, the depths cancel exactly and no correction from Out is needed.

The argument is quadratic and translation invariant, and both features are lost at $p > 2$; see Proposition 4.1.

3. The case S_4

3.1. **The norm.** The S_4 norm is as explicit as the S_2 norm.

Lemma 3.1. *For φ with $\widehat{\varphi} \in L^2(2D)$,*

$$\|H_\varphi\|_{S_4}^4 = \iiint \widehat{\varphi}(u)\widehat{\varphi}(w)\overline{\widehat{\varphi}(v)}\,\overline{\widehat{\varphi}(u+w-v)}\,\Omega(u,v,w)\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w,$$

where $\Omega(u,v,w) = |D \cap (u-D) \cap (v-D) \cap (v-w+D)|$.

Proof. Let $K(\xi,\eta) = \widehat{\varphi}(\xi+\eta)\chi_D(\xi)\chi_D(\eta)$. Then

$$\|H_\varphi\|_{S_4}^4 = \operatorname{tr}((H^*H)^2) = \int_{D^4} \overline{K(a,b)}K(a,d)\overline{K(c,d)}K(c,b)\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}c\,\mathrm{d}d.$$

Substitute $u = a+d$, $v = a+b$, $w = c+b$, so that $c+d = u+w-v$, and integrate out a: the remaining constraints are $a \in D$, $v-a \in D$, $u-a \in D$ and $a+w-v \in D$. □

So (N_4) is an inequality of Helson type of order four: $|\langle h,\varphi\rangle|^4 \le C\|h\|_1^4$ times the quartic form of Lemma 3.1. Unlike (2) it is not a weighted L^2 estimate for $\widehat{h}$, and duality does not turn it into a quadratic form in h.

3.2. **Why four: the lower bound.** The threshold 4 appears on both sides of (N_4). We first recast Bampouras's counterexample as a counting argument in which the exponent splits as $4 = 2(1+\gamma)$.

Lemma 3.2. *Let $\widehat{b}$ be supported in the unit ball of $\mathbb{R}^n$, let $\rho > 0$, and let $|\xi_i - \xi_j| \ge 2\rho$ for $i \ne j$. For every integer $k > n/4$,*

$$\Big\|\sum_j c_j e^{2\pi i\xi_j\cdot x}\rho^n b(\rho x)\Big\|_1 \le \|(1+|x|^2)^k b\|_2 \|(1+|x|^2)^{-k}\|_2 \|c\|_{\ell^2}.$$

Proof. Let f be the sum and $w(x) = (1+|\rho x|^2)^k$. By Cauchy–Schwarz, $\|f\|_1 \le \|wf\|_2\|1/w\|_2$, and $\|1/w\|_2 = \rho^{-n/2}\|(1+|x|^2)^{-k}\|_2$. Multiplication by w is a differential operator on the Fourier side, so the terms of wf keep their disjoint Fourier supports $B(\xi_j,\rho)$. Each has L^2 norm $|c_j|\rho^{n/2}\|(1+|x|^2)^k b\|_2$, and Plancherel finishes. □

Let $\beta \in C_c^\infty(B(0,1))$ be nonzero and let $z_1,\dots,z_N \in 2D$ with $B(z_i,\rho) \subset 2D$, $|z_i - z_j| \ge 2\rho$, and the lenses $D \cap (B(z_i,\rho) - D)$ pairwise disjoint. Put $\widehat{\psi} = \sum_i \beta((\cdot - z_i)/\rho)$ and $a = \max_i \sup_{B(z_i,\rho)} \omega$.

Proposition 3.1 (Bampouras [1])**.** *For $p \ge 2$,*

$$\frac{\langle\psi,\psi\rangle}{\|\psi\|_1\|H_\psi\|_{S_p}} \ge c(\beta,p)\,N^{1/2-1/p}\Big(\frac{\rho^2}{a}\Big)^{1/p}.$$

Proof. The pairing equals $N\rho^2\|\beta\|_2^2$, and Lemma 3.2 gives $\|\psi\|_1 \le C\sqrt{N}$, independently of ρ. The Hankel operators of the individual caps act on the orthogonal subspaces L^2 of the lenses, so Bampouras's bound $\|H_\varphi\|_{S_p} \le \|\widehat{\varphi}\,\omega^{1/p}\|_{p'}$ [1] gives $\|H_\psi\|_{S_p}^p \le N(\|\beta\|_{p'}\rho^{2/p'}a^{1/p})^p$. □

Caps of radius $\rho \asymp \varepsilon^2$ at distance $\asymp \varepsilon^2$ from $\partial(2D)$, spaced along an arc, satisfy the hypotheses with $N \asymp \varepsilon^{-1}$ and $a \asymp \varepsilon^3$ [1]. Then $\rho^2/a \asymp N^{-1}$ and the quotient is $\gtrsim N^{1/2-2/p}$, which is unbounded exactly when $p > 4$. In general, if $\rho^n/a \asymp N^{-\gamma}$ the threshold is $p = 2(1+\gamma)$. The factor 2 is square-root cancellation in L^1 (Lemma 3.2 is a Khintchine inequality without randomness), and $\gamma = 1$ is parabolic scaling at a curved boundary; for $N \asymp \varepsilon^{-k}$ caps in $\mathbb{R}^n$

the same count gives the thresholds $2(n+k-1)/k$, which equal 4 for $k=n-1$. For polygons no such family exists, consistent with [5]. At $p=4$ the lower bound stops growing.

On the infinite polydisc the threshold has a different source. There $\|H_\psi\| \geq \|H_\psi 1\| = \|\psi\|_2$ for every symbol, so sums of symbols never split into orthogonal blocks, and the 4 of [2] comes from the fourth moment of a sum of independent Steinhaus variables, $\mathbb{E}|Z|^4 = 2$ in the Gaussian limit.

3.3. **Why four: tube packing.** Expand a symbol in wave packets, $\varphi = \sum_T b_T \psi_T$. The S_p norm controls $\|b\|_{\ell^p}$ by the necessary Besov estimate [4, Theorem 4.3]; see Section 4. The natural route from ℓ^p coefficients to a pairing with PW^1 is through a Carleson condition, as for product BMO, and this requires a packing estimate for tubes. Write $\mathcal{T}_k$ for the tubes of depth 4^{-k}, and fix k_0 with $4^{k_0} \geq 64$.

Lemma 3.3 (Tube packing)**.** *For every measurable $U \subset \mathbb{R}^2$ of finite measure,*

$$\sum_{k \geq k_0} \sum_{T \in \mathcal{T}_k,\, T \subset U} |T|^2 \leq 240 |U|^2.$$

Proof. Let $n_k(x)$ be the number of directions at depth k whose tile through x lies in U. Tiles of one direction overlap only on boundaries, so the left side equals $\sum_k |T_k| \int_U n_k$, and it suffices to show $\sum_k n_k(x)|T_k| \leq 240|U|$ for $x \in U$. Let $T \subset U$ contain x. One end of T is at axial distance at least $L_k/2$ from x; let $A(T)$ be the part of T at axial distance between $L_k/4$ and $L_k/2$ from x on that side. Then $|A(T)| = |T|/4$ and $A(T)$ lies in the annulus $R_k(x) = \{L_k/4 \leq |y-x| \leq 0.52 L_k\}$; these annuli are disjoint for different k. A point $y \in A(T)$ sees the axis of T under an angle at most $\arcsin(4W_k/L_k) \leq 2\pi\sqrt{\delta_k}$ from $\pm(y-x)$, so at most $2(4\pi+1) \leq 60$ of the sets $A(T)$, one per direction, contain y. Hence $n_k(x)|T_k|/4 \leq 60 |U \cap R_k(x)|$, and we sum over k. □

By Cauchy–Schwarz, Lemma 3.3 gives the Carleson bound $\sum_{T \subset U} |b_T|^2 |T| \leq \sqrt{240}\, \|b\|_{\ell^4}^2 |U|$. With ℓ^p coefficients the same argument needs $\sum_{T \subset U} |T|^s \lesssim |U|^s$ with $s = p/(p-2)$. For $U = B(0,R)$ the tubes inside U have depth $\delta \geq 1/R$, there are $\asymp \delta^{-1/2}$ directions and $\asymp R^2 \delta^{3/2}$ tubes per direction, so

$$\sum_{T \subset B(0,R)} |T|^s \asymp R^2 \sum_{\delta \geq 1/R} \delta^{1-3s/2} \asymp R^{1+3s/2},$$

which is $\lesssim R^{2s}$ if and only if $s \geq 2$, that is $p \leq 4$. Heuristically, the saturating coefficients, constant on all tubes of depth $1/R$ in the ball, are the packet expansion of Bampouras's caps, so the counterexample above 4 and the packing estimate below 4 are two readings of the same configuration.

The same exponent appears in an embedding of square functions into packet masses. A tube T is *half inside* Ω if $|T \cap \Omega| \geq |T|/2$.

Lemma 3.4. *For every measurable Ω of finite measure, $\sum |T|^2 \leq 3600 |\Omega|^2$, the sum taken over the tubes of depth at most 4^{-k_0} half inside Ω.*

Proof. As for Lemma 3.3, now with $\sum |T| \leq 2 \sum |T \cap \Omega|$, and with $A(T)$ replaced by the part of T at axial distance at least $L_k/8$ from x, which carries at least a quarter of $|T|$ inside Ω; these parts lie in $\{L_k/8 \leq |y-x| \leq 1.02 L_k\}$, overlap at most $2(16\pi+1) \leq 220$ times, and the annuli are disjoint for depths of equal parity. □

Theorem 3.1 (Tent embedding)**.** *Let $a_T \geq 0$ be coefficients on tubes of depth at most 4^{-k_0} and $S = (\sum_T a_T^2 \mathbf{1}_T)^{1/2}$. Then $\sum_T (|T| a_T)^{4/3} \leq C (\int S)^{4/3}$, and the exponent $4/3$ cannot be lowered.*

Proof. Let $\Omega_j = \{S > 2^j\}$, so $\sum_j 2^j |\Omega_j| \leq 2 \int S$. Since $S \geq a_T$ on T, the index $j(T) = \max\{j : T \text{ half inside } \Omega_j\}$ is defined when $a_T > 0$. Put $\lambda_T = |T|^{-1} \int_T S^{-1}$. More than half of T lies

where $S \le 2^{j(T)+1}$, so $\lambda_T^{-1} \le 2^{j(T)+2}$. Now $\sum_T |T| a_T^2 \lambda_T = \int S^{-1} \sum_T a_T^2 \mathbf{1}_T = \int S$, and by Lemma 3.4

$$\sum_T |T|^2 \lambda_T^{-2} \le 16 \sum_j 4^j \sum_{j(T)=j} |T|^2 \le C \sum_j (2^j |\Omega_j|)^2 \le C \Big(\int S \Big)^2 .$$

Hölder's inequality with exponents $3/2$ and 3 applied to $(|T| a_T)^{4/3} = (|T| a_T^2 \lambda_T)^{2/3} (|T|^2 \lambda_T^{-2})^{1/3}$ gives the bound. For sharpness let $a_T = 1$ on the $\asymp R$ tubes of depth $1/R$ in $B(0,R)$: then $\sum (|T| a_T)^q \asymp R^{1+3q/2}$ and $\int S \asymp R^{9/4}$, which forces $q \ge 4/3$. □

4. Formulations

4.1. **Besov and packet forms.** For a smooth parabolic Whitney partition $\sum_Q m_Q = 1$ of $2D$ (with finitely many interior pieces) put $h_Q = \mathcal{F}^{-1}(m_Q \widehat{h})$ and

$$\mathcal{B}_p(h) = \Big(\sum_Q (V_Q^{-1/p} \|h_Q\|_{p'})^{p'} \Big)^{1/p'} . \tag{E_p}$$

By [6, Lemma 7.2], which uses only the necessary Besov estimate and holds for every $1 \le p \le \infty$, the embedding $\mathcal{B}_p(h) \lesssim \|h\|_1$ on $\mathrm{PW}^1(2D)$ implies (N_p). For $p < 4$ the converse holds as well, by duality with the Besov characterization of [4]. At $p = 4$ the embedding reads $\sum_Q \delta_Q^{-1/2} \|h_Q\|_{4/3}^{4/3} \lesssim \|h\|_1^{4/3}$.

Since h_Q is essentially constant in modulus on the tubes dual to Q, Bernstein's inequality gives, up to Schwartz tails, $\|h_Q\|_{p'}^{p'} \asymp \sum_T |T|^{1-p'} m_T(h)^{p'}$, and as $V_Q^{-p'/p} |T|^{1-p'} \asymp 1$, ($\mathrm{E}_p$) is equivalent to the *packet-mass inequality*

$$\sum_T m_T(h)^{p'} \lesssim \|h\|_1^{p'} . \tag{3}$$

For $p' = 2$ this is Helson's inequality (2); for $p' = 1$ it would be Nehari's theorem for bounded operators, which is false [8]. The endpoint is $p' = 4/3$. By (1), (3) is equivalent to the *dual packet inequality*

$$\operatorname{dist}_{L^\infty} \Big(\sum_T b_T \psi_T, \mathrm{Out} \Big) \lesssim \|b\|_{\ell^p} \qquad \text{for finitely supported } b, \tag{4}$$

which says that ℓ^p sums of disc wave packets are, modulo functions with spectrum outside the disc, bounded.

4.2. **Weighted Helson inequalities.** The S_4 problem is a Helson inequality with weights that depend on h.

Proposition 4.1. *For every h,*

$$\Big(\sum_T m_T(h)^{4/3} \Big)^{3/2} = \inf \Big\{ \sum_T m_T(h)^2 w_T : \ w_T > 0, \ \sum_T w_T^{-2} \le 1 \Big\} .$$

Proof. Hölder's inequality with exponents $3/2$ and 3 applied to $m_T^{4/3} = (m_T^2 w_T)^{2/3} w_T^{-2/3}$ gives $\le$. For finitely many nonzero m_T, equality holds for $w_T = m_T^{-2/3} (\sum m^{4/3})^{1/2}$ when all $m_T > 0$; tubes with $m_T = 0$ get large weights, which change both sides arbitrarily little. □

So (N_4) follows if for every h there are weights with $\sum_T w_T^{-2} \le 1$ and $\sum_T m_T(h)^2 w_T \lesssim \|h\|_1^2$. Helson's inequality is the case of constant weights, which are not admissible since $\sum_T w_T^{-2} = \infty$. In the form of Remark 2.1, $\sum_T m_T^2 w_T = \langle K_w h, h \rangle$ with $K_w(x,y) = \sum_T w_T \overline{\psi_T(x)} \psi_T(y)$, and the S_2 proof would extend if K_w were bounded modulo Out in each variable. For admissible weights K_w is not translation invariant, and the Fourier positivity behind Proposition 2.1 is no longer available. The proof of Theorem 3.1 is an instance of Proposition 4.1: it uses $w_T \propto \lambda_T / |T|$, the average of S^{-1} over T.

4.3. **Square functions.** Let $a_T = m_T(h)/|T|$ and $S_h = (\sum_T a_T^2 \mathbf{1}_T)^{1/2}$, the parabolic square function of h (all depths and directions). By Theorem 3.1, the endpoint follows from

$$\int_{\mathbb{R}^2} S_h \lesssim \|h\|_1, \qquad h \in \mathrm{PW}^1(2D). \tag{B}$$

This is an L^1 estimate for a Littlewood–Paley square function adapted to the boundary of the disc. It cannot hold for general L^1 functions: for a bump at the origin with spectrum not inside $2D$, every tube through the origin at every depth carries a packet mass of order one. The constraint $\widehat{h} = 0$ outside $2D$ is used in an essential way. At a single depth δ, for h with spectrum in the annulus $\{2 - |\xi| \asymp \delta\}$, (B) is a square function estimate for the decomposition of a thin annulus into caps of angle $\sqrt{\delta}$; there the Out-freedom is replaced by freedom off the annulus.

By Hölder's inequality, single-depth (B) follows from the weighted pair

$$\int |h|^2 S^{-1} \gtrsim \int S, \qquad \int |h|^4 S^{-3} \lesssim \int S, \tag{5}$$

since $\int |h|^2 S^{-1} \le (\int |h|)^{2/3} (\int |h|^4 S^{-3})^{1/3}$. When S is roughly constant on balls of radius L, both follow from orthogonality and the Córdoba–Fefferman biorthogonality of the circle on L-balls. A weaker sufficient condition replaces the second inequality by a logarithmic moment: by convexity of $p \mapsto \log \int (|h|/S)^p S$,

$$\|h\|_1 \ge \int |h|^2 S^{-1} \cdot \exp\Big(- \frac{\int |h|^2 S^{-1} \log(|h|/S)}{\int |h|^2 S^{-1}} \Big),$$

so it suffices that $\int |h|^2 S^{-1} \log^+(|h|/S) \lesssim \int S$.

4.4. **The range $p < 4$.** Logarithmic losses in the number of packets are harmless below 4.

Proposition 4.2. *Suppose that, for $\|h\|_1 = 1$, $\#\{T : m_T(h) > \lambda\} \le C\lambda^{-4/3} \log^A(2/\lambda)$ for $0 < \lambda \le 1$. Then (3) holds for every $p' > 4/3$, and Nehari's theorem holds in S_p for every $p < 4$.*

Proof. $m_T(h) \le \|\psi_T\|_\infty \|h\|_1 = 1$, so the layer-cake formula gives $\sum_T m_T^q = q \int_0^1 \lambda^{q-1} \#\{m_T > \lambda\} \, \mathrm{d}\lambda < \infty$ for $q > 4/3$. Apply [6, Lemma 7.2]. □

The weak bound reduces to a geometric count and a cancellation estimate. Fix λ and a finite set F of n tubes with $m_T > \lambda$, let F_δ be those of depth δ, $n_\delta = \#F_\delta$, $\mu_\delta = \sum_{T \in F_\delta} \mathbf{1}_T$, and $M_T = 1 + |T|^{-1} \int_T \mu_\delta$ for $T \in F_\delta$.

Proposition 4.3. *At each depth, $\max_{F_\delta} M_T \le C\sqrt{n_\delta}$, with C absolute.*

Proof. Fix T and group the other directions by their angle α to T into bins $j/W \le \alpha < (j+1)/W$, each containing $O(1)$ directions. A tube in bin j meets T in area at most $C|T|/(1+j)$, and at most $C(1+j)$ tubes of one direction in bin j meet T. If c_j selected tubes in bin j meet T, then $c_j \le C(1+j)$ and $\sum_j c_j \le n_\delta$, so for every $J \ge 1$, $M_T \le 1 + C\sum_j c_j/(1+j) \le C(1 + J + n_\delta/J)$. Take $J = \lceil \sqrt{n_\delta} \rceil$. □

For each depth put $\Phi_\delta = \sum_{T \in F_\delta} \varepsilon_T M_T^{-1/2} \psi_T$ with ε_T the phase of $\langle h, \psi_T \rangle$. By Proposition 4.3, $\sum_{F_\delta} M_T^{-1/2} \ge c n_\delta^{3/4}$, so

$$c\lambda n^{3/4} \le \lambda \sum_\delta n_\delta^{3/4} \le \Big\langle h, \sum_\delta \Phi_\delta \Big\rangle \le \mathrm{dist}_{L^\infty}\Big(\sum_\delta \Phi_\delta, \mathrm{Out} \Big).$$

Together with Helson's inequality, which gives $n \lesssim \lambda^{-2}$, this proves the following.

Corollary 4.1. *If, for all finite tube families and all phases,*

$$\mathrm{dist}_{L^\infty}\Big(\sum_\delta \Phi_\delta, \mathrm{Out} \Big) \le C \log^a(2 + n), \tag{C}$$

then Nehari's theorem holds in $S_p(\mathrm{PW}^2(D))$ for every $p < 4$.

The normalization by multiplicity must be taken depth by depth: shallow tubes packed in a ball together with the deep tubes containing that ball have $\sum M_T \asymp n^2$ if multiplicities are counted across depths. Estimate (C) states that packet sums with arbitrary phases, normalized by multiplicity, exhibit square-root cancellation modulo Out. Remark 2.1 is evidence that combining depths helps rather than hurts: the complete bush through a point is bounded without any normalization.

5. Tools that use the disc

The only mechanism we know for using Out at a single depth is to push a packet's spectrum across the circle. Fix a depth δ, a Schwartz profile b with $\widehat{b}$ supported in $[-\varepsilon,\varepsilon]^2$, $\varepsilon = 10^{-3}$, and packets

$$\psi(x) = e^{2\pi i\xi\cdot x} b\big(\delta(x-t)\cdot\theta, \sqrt{\delta}(x-t)\cdot\theta^{\perp}\big), \qquad \xi = 2(1-10\delta)\theta.$$

Lemma 5.1 (Outward modulation)**.** *For every $a\in\mathbb{R}^2$, $\psi(x)e^{2\pi i\,40\delta\theta\cdot(x-a)}\in$ Out. Hence ψ and $\psi^a = \psi\,(1-e^{2\pi i\,40\delta\theta\cdot(x-a)})$ have the same pairing with every $h\in \mathrm{PW}^1(2D)$.*

Proof. The modulated spectrum has component along θ at least $2+20\delta-\varepsilon\delta > 2$, and $|v|\ge v\cdot\theta$. □

More generally $\psi\equiv\psi(1-e^{2\pi i\,40\delta\theta\cdot x}G)$ modulo Out whenever $G\in L^\infty$ has spectrum in the half plane $\{\eta\cdot\theta\ge 0\}$, for instance a function of $x\cdot\theta^\perp$ times a bounded analytic function of $x\cdot\theta$. The factor ψ^a is small near the line through a perpendicular to θ: $|\psi^a(x)|\le 80\pi\delta|(x-a)\cdot\theta||\psi(x)|$.

Proposition 5.1 (Anchored bushes)**.** *Let $\mathcal{F}$ be a family of packets whose tubes contain a common point a, with $\sqrt{\delta}$-separated directions, and let $|\beta_\psi|\le\beta$. Then $\sup_x|\sum_{\mathcal{F}}\beta_\psi\psi^a(x)|\le C\beta$, with C depending only on b.*

Proof. Let $\rho=|x-a|$. If $\rho\le W$, there are at most CW terms, each at most $C\beta\delta W$. If $W<\rho\le L$, the packets whose axis passes at distance between jW and $(j+1)W$ from x have directions in an arc of length $C(1+j)W/\rho$, so they number at most $C(1+j)L/\rho$, and each is bounded by $C_N(1+j)^{-N}$; hence $\sum|\psi(x)|\le CL/\rho$, and the factor $80\pi\delta\rho$ gives $C\beta$. If $\rho>L$, the decay of b in both variables absorbs the count. □

Without the correction, a bush of $N\asymp W$ packets focused at a has $|\sum\beta_\psi\psi|\asymp N\beta$ near a; with it, the whole bush, including its Bessel-type tail, costs $O(\beta)$. In (C) a family through one point therefore costs $O(\max M_T^{-1/2})$. Compare Remark 2.1, where the complete bush over all depths needs no correction. Heuristically, a bounded analytic function of $x\cdot\theta$ with prescribed zeros (a Blaschke product) suppresses a packet near several points of its axis, at a cost $e^{O(\delta\sum_j\rho_j)}$ for suppression on intervals of lengths ρ_j, so each tube can afford suppression on a set of total length $O(L)$ at bounded cost.

6. What a proof must take into account

The following features are, as far as we can see, unavoidable.

The constant is not 1. For every $p\ge 2$ there are h,φ with spectrum in $2\overline{D}$ and $|\langle h,\varphi\rangle| > \|h\|_1\|H_\varphi\|_{S_p}$. Indeed $h_0=\check{\chi}_D^2\ge 0$ has $h_0(0)=|D|^2=|D|\|h_0\|_1$; adding $\varepsilon\check{\beta}$ with $\beta\ge 0$ supported in $2D$ away from 0 raises $h(0)$ by order ε while changing $\|h\|_1$ by $o(\varepsilon)$; and symbols with $\widehat{\varphi}\uparrow\mathbf{1}_{2D}$ have $\|H_\varphi\|_{S_p}\to|D|$ and $\langle h,\varphi\rangle\to h(0)$. Tensor powers show that the constant on $\mathrm{PW}(D^m)$ grows exponentially in m. In particular no argument that is exact for a single symmetric configuration can be tight.

Depth logarithms are not free. Proposition 4.2 allows losses in $\log n$ or $\log(1/\lambda)$, but not in $\log(1/\delta)$: a single L^1-normalized packet of depth δ has packet mass $\asymp 1$ at every δ, so a coefficient threshold does not bound the depth. Induction on angular scales of Bourgain–Guth type costs a constant per scale and hence a power of $\log(1/\delta)$, unless the number of costly scales is bounded in terms of n.

Focusing occurs at unit scale. At one depth, $\int|h|^2 = \int S^2$ by orthogonality, so $\|h\|_1$ can be small compared with $\int S$ only if L^2 mass concentrates. A bush of N packets aligned at a point focuses on a unit ball, but it is harmless: $|h| \asymp S \asymp W|x|^{-1/2}$ on most of the bush, so $\|h\|_1 \asymp \int S$. Even so, at $p = 4$ the bush carries a logarithm in L^4, so arguments through L^4 must be lossless. The more dangerous configurations are m adjacent directions with aligned phases: they focus on a Dirichlet strip of width W/m and length L/m^2 (the curvature of the circle limits the length), where $|h| \asymp m$ and $S \asymp \sqrt{m}$ times the amplitude. The L^4 excess of the strip is comparable to the S^4 mass of the overlap region of the m tubes, of size $W \times L/m$, so local estimates at any scale finer than L/m fail, while the global Córdoba–Fefferman bound is sharp.

Foci recur. It is tempting to bound the number of foci a tube takes part in. This fails: with $W = 256M^4$ and $D = 256M^5$, the M packets with exact frequencies $\xi_j = (r - j/D, (r^2 - (r - j/D)^2)^{1/2})$, $|\xi_j| = r$, satisfy $\xi_j \cdot (kD, 0) = rkD - jk$, so they are in phase at all $M + 1$ points $(kD, 0)$, $0 \le k \le M$, which lie inside every tube. A proof must count the mass of focal regions, not their number.

Carving (heuristic). The weight S^{-3} in (5) depends on all packets. Transversal tubes crossing the overlap region of a narrow family raise S off its Dirichlet strip but not on it. Then the narrow family's contribution to $\int|h|^4S^{-3}$ is of order $m^{-1/2}$ times its amplitude times WL, which is not controlled by the narrow family's own $\int S^4S^{-3}$, but is controlled by $\int S$ over its overlap region, including the transversal mass. So an induction on subfamilies of directions with the ambient weight does not seem able to prove (5); the excess of one family must be charged to the square function of the families crossing it. Charging the excess of each narrow family τ to $\int_{R_\tau} S$ over its overlap region R_τ costs the S-weighted average of $(\#\{\tau : x \in R_\tau\})^{1/2}$, a Kakeya multiplicity of the rectangles R_τ. At a single angular scale this is at most a power of $\log n$. Across scales it is not: short rectangles form bushes of multiplicity comparable to their eccentricity. Bushes, however, produce no excess. The charging must therefore use only the families that actually produce excess.

Corrections must cancel. Lemma 5.1 and Blaschke factors suppress each packet on any prescribed set of length $O(L)$ along its axis. Absolute bounds on the resulting tails, summed over packets, are polynomially large in n, so the corrections of different packets must themselves cancel, as the depths do in Remark 2.1.

7. The missing estimate

In the form closest to the available tools, the problem is the following.

Conjecture 7.1. *There is C such that $\int_{\mathbb{R}^2} S_h \le C\|h\|_1$ for every $h \in \mathrm{PW}^1(2D)$.*

By Theorem 3.1 and [6, Lemma 7.2], Conjecture 7.1 implies Nehari's theorem in $S_4(\mathrm{PW}^2(D))$, and hence in S_p for $p \le 4$; it also implies the dual packet inequality (4) at $p = 4$. Its single-depth case, uniformly in the depth, is open as well; a version with losses polynomial in $\log n$ in the form (C), uniformly in the depths, would give every $p < 4$ by Corollary 4.1. In the language of Section 2, the S_2 proof uses one bounded function, the complete bush $e^{4\pi i|x|}$; Proposition 4.1 asks for a bounded bush adapted to h. We expect a proof to combine a decomposition into broad and narrow points of high multiplicity across angular scales, the lossless Córdoba–Fefferman bound for each narrow family on its own overlap region, and a single global charging step whose Kakeya-type multiplicity is controlled by the number of excess-producing families. The disc constraint should enter through Lemma 5.1 and through the cancellation between depths seen in Remark 2.1.

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